

Cross Border Car Hailing - App Summary

Cross-border car booking for foreign travelers moving between Hong Kong and Mainland China.
Two mobile apps (customer + driver) backed by a reservation-first marketplace, top-5 driver choice, Stripe payment in HKD, and admin/ops controls.

1. Product info

Item	Value
Target customer	Foreign travelers, business visitors, expats, non-local customers between HK and Mainland China.
v1 routes	HK Airport ↔ Shenzhen, HK City ↔ Shenzhen, HK ↔ Guangzhou (both directions).
Service coverage	All HK areas; all Shenzhen and Guangzhou areas.
Booking model (v1)	Reservation-first marketplace with top-5 driver choices.
Driver supply	Open marketplace — individual drivers and fleet/company accounts.
Mobile stack	Kotlin Multiplatform (KMP), native UI per platform.
Map/routing	AMap (authoritative for HK + Mainland routing and fare distance).
Payment	Stripe credit card, HKD, HK merchant entity.
Driver payout	Weekly bank transfer outside Stripe.
Support hours	In-app chat 09:00–00:00, target response < 5 min on active trips.
Languages	English, Traditional Chinese, Simplified Chinese.

2. Roles

- **Customer** — search, book, pay, track, chat, review.
- **Driver** — verify, set availability, submit offers, execute trips, get paid.
- **Admin/Ops** — verify drivers, configure pricing/areas, monitor bookings, override.
- **Support agent** — joins chat by request, handles cancellations, refunds, emergencies.

3. Core app behaviour

Customer journey (reservation, v1)

1. Choose language and contact method.
2. Enter pickup, dropoff, date/time, passengers, luggage, vehicle class, special needs.

3. System validates route support and shows reference fare, included/excluded fees, cancellation policy, offer expiry.
4. Eligible marketplace drivers have **2 minutes** to submit offers.
5. System ranks offers by balanced value and shows up to **top 5** choices (fewer if not enough offers).
6. Customer selects one driver — or rejects all and creates a new request.
7. Customer pays the **full fare** by Stripe credit card in HKD.
8. Booking confirmed only after payment succeeds.
9. Customer tracks driver: en route → arrived → trip started → border crossing → completed.
10. Receipt, optional review, support ticket if needed.

Driver journey

1. Complete identity, vehicle, cross-border, insurance, business document verification.
2. Set availability and service routes; backend also checks **current area**.
3. Receive eligible reservation request (filtered before display).
4. Submit offer (own price) or decline within the 2-minute window.
5. If selected, receive confirmed job and customer execution info (only after payment succeeds).
6. Navigate to pickup → mark arrived → start trip → update border crossing → complete dropoff.
7. View payout state; settled weekly by bank transfer.

4. Functional spec

4.1 Booking models

- **Reservation (v1)** — advance booking, full charge at booking. Reservation fare = `instant_reference_fare × 0.8` (20% discount).
- **Instant (post-MVP)** — Stripe authorization hold + manual capture after ride.
- **Bidding (filtered)** — drivers set own price; platform filters by eligibility/quality and shows top 5 only.

4.2 Top-5 ranking

Hard eligibility filters first, then weighted scoring:

Signal	Weight
Driver reliability & completed trips	30%
Price/value vs reference fare	25%
Vehicle fit & service class	20%
Availability/wait confidence	15%
Language/support fit	10%

Backend (not the app) generates the visible set. Each visible choice shows: final price, vehicle class + photo, driver rating + completed trip count, pickup readiness, included/excluded fees, luggage/passenger

fit, language, cancellation summary.

4.3 Driver eligibility

A driver must be:

- Account, vehicle, and cross-border documents verified.
- Direction-capable and **currently in the pickup-side area**.
- Border-route capable.
- Capacity-fit for passengers + luggage.
- Available, no conflicting booking.
- Above rating/quality threshold.
- Not under temporary ban.

4.4 Fare engine (configurable per route/class/time)

```
fare =  
  base_fare  
  + distance_km * per_km_rate  
  + estimated_time_min * per_min_rate  
  + tolls_and_route_fees  
  + border_service_fee  
  + vehicle_class_fee  
  + luggage_or_special_request_fee  
  + demand_adjustment  
  - reservation_discount
```

Distance is authoritative from AMap.

4.5 Payment rules

Event	Action
Reservation: customer selects driver	Create Stripe PaymentIntent (HKD), capture immediately.
Reservation: payment succeeds	Booking → <code>payment_paid</code> , locked to driver.
Instant: customer confirms	Create manual-capture PaymentIntent, authorize.
Instant: trip completes	Capture authorized amount.
Reservation: cancel > 24h before pickup	Refund 85%, keep 15% fee.
Reservation: cancel < 24h before pickup	Open decision.
Reservation: cancel after driver arrival	Open decision.
Instant: cancel before driver arrival	Cancel PaymentIntent, release hold.
Instant: cancel > 5 min after driver arrived	Capture 10% fee, release 90%.
Driver cancels (no replacement)	Full refund / support replacement; log + penalty.
Trip completes	Move driver payout to weekly bank transfer.

4.6 Driver cancellation policy

- ≤ 5 min after accept → allowed but logged.
- 5 min after accept → warning/penalty record.
- 3 late cancellations in rolling week → 5-hour temporary ban.

4.7 Booking state machine (backend-owned)

```
draft → quote_generated → collecting_driver_offers → offers_ready
  → customer_selecting_driver → driver_selected → payment_pending
  → payment_paid (or payment_authorized for instant)
  → driver_confirmed → customer_confirmed
  → driver_en_route → driver_arrived → trip_started
  → border_crossing → trip_completed → settlement_pending → settled
Side states: quote_expired, payment_failed,
  cancelled_by_customer / driver / admin, refunded, disputed
```

4.8 Payment state machine

```
payment_not_started → payment_processing → payment_requires_action
  → payment_authorized | payment_paid → payment_captured | payment_released
Side states: refund_requested, refund_review_required,
  refund_partial_85_percent, refund_full,
  cancellation_fee_retained_15_percent,
  instant_late_cancel_fee_captured_10_percent,
  payment_failed, payment_disputed
```

4.9 Customer info visibility to driver

- **Before** customer accepts and pays: only operational booking details. **No** nationality, no language.
- **After** payment succeeds: full execution info (name, contact, pickup notes).

4.10 Support and chat

- In-app customer ↔ driver chat, support joins by help request.
- Button-triggered auto-translation.
- Operating hours 09:00–00:00; active-trip response target < 5 minutes.
- Emergency contact workflow.

4.11 Admin capabilities (v1)

Driver/vehicle verification, document expiry tracking, fare rule editor, service area editor, booking timeline, marketplace offer monitoring, top-5 override/suppression, payment/refund override, support tickets, driver quality flags, audit log.

5. Feature checklist

Customer app

- ☐ Language + contact onboarding (EN / TC / SC, international phone).
- ☐ Route search with service-area validation.
- ☐ Fare quote with included/excluded fees + offer expiry.
- ☐ Reservation booking form (passengers, luggage, vehicle class, special needs).
- ☐ 2-minute offer collection with progress indicator.
- ☐ Top-5 driver choices view (compare price, rating, vehicle, fees, availability).
- ☐ Select one driver / reject all + restart.
- ☐ Stripe credit-card payment in HKD.
- ☐ Live trip tracking (en route, arrived, started, border, completed).
- ☐ In-app chat with auto-translate button + support join.
- ☐ Booking history, receipts, ratings.

- ☐ Cancellation flow with policy display.

Driver app

- ☐ Onboarding with full document set (identity, license, vehicle, permit, insurance, business).
- ☐ Availability + service zone + current-area state.
- ☐ Filtered eligible job feed.
- ☐ Offer submission (price, included fees, availability).
- ☐ Accept/decline + 5-minute grace window.
- ☐ Confirmed-job detail post-payment.
- ☐ Navigation handoff (AMap native).
- ☐ Trip state updates (arrived, start, border, complete).
- ☐ Earnings + weekly payout view.
- ☐ Issue reporting (no-show, route, border delay, payment).
- ☐ Document expiry alerts.

Admin/ops dashboard

- ☐ Verification queues (driver, vehicle, documents).
- ☐ Document expiry dashboard.
- ☐ Booking search + timeline.
- ☐ Marketplace offer monitoring + top-5 override.
- ☐ Fare + service area editors.
- ☐ Refund/cancellation override + Stripe PaymentIntent view.
- ☐ Support ticket queue.
- ☐ Driver quality flags + temporary ban controls.
- ☐ Audit log.

6. Non-functional + architecture

- **Mobile** — KMP shared modules: `auth`, `booking`, `pricing`, `driver-offers`, `trip-state`, `profile`, `payments`, `notifications`, `localization`, `map-provider`, `api-client`. Native UI + native AMap SDK per platform.
- **Backend modules** — `auth`, `customer`, `driver`, `vehicle/document`, `booking`, `pricing`, `dispatch`, `offer-ranking`, `payment`, `trip-tracking`, `notification`, `support`, `admin`, `audit`. Runtime leaning Ktor (Kotlin) to share the domain module with KMP — to confirm; see tech-decisions.
- **Data store** — PostgreSQL + PostGIS (PostGIS for service-area, route, and location queries).
- **Cloud (v1)** — HK / global hosting is acceptable for v1. Mainland deployment only if legal/network/distribution forces it. Hybrid kept as an option. Server hosting location and iOS/Android app distribution (incl. possible ICP / 备案) are still open — see tech-decisions.
- **Business rules in backend config**, not mobile — fare rates, route support, vehicle classes, reservation discount, ranking weights, Stripe rules, cancellation rules, driver eligibility, border

crossings, map provider selection, notification templates.

7. Out of MVP

Fully automatic instant dispatch, showing every bid to customer, real-time price negotiation, complex surge, corporate billing, loyalty program, multi-stop routes, in-app wallet, advanced fraud detection, fully automated compliance decisioning.

8. Open decisions still blocking implementation

- Cancellation/refund rule for customer cancel **within 24h** before pickup.
- Cancellation/refund rule for **reservation after-arrival** cancellation.
- Bank-transfer process for weekly driver payout.
- Address normalization across EN / TC / SC.
- Final legal naming for required cross-border permits.
- **Infrastructure decisions** — server hosting location, iOS/Android app distribution, ICP / 备案, China entity, backend runtime, and v1 real-time scope. Tracked in tech-decisions.

9. Related notes

- [_index](#)
- [teammate-overview](#)
- [product-logic](#)
- [interfaces](#)
- [mvp](#)
- [data-model](#)
- [payment-and-cancellation](#)
- [architecture-and-cloud](#)
- [tech-decisions](#)
- [open-questions](#)